KRE Study 3 — When do falling Treasury yields help or hurt regional banks? (with the rate-beta flip)

Data through the 10/30/2026 close · prepared 11/02/2026

Bottom line

1. **Falling yields have not been a reliable buy signal for KRE.** Since June 2006, falling-yield windows were followed by *slightly below-baseline* 60-session KRE returns (−0.2% to +0.7% vs a +2.0% baseline; not significant). The one result that holds across window lengths, thresholds, and the removal of the COVID and SVB periods is relative: **after the 2-year yield falls, KRE has tended to lag KBWB over the next 60 sessions** (−1.9% to −2.2% vs a −0.5% baseline; Welch t ≈ −2.3 to −3.0; 25 windows in each grid, 14–15 separate episodes, 2012–2026 only). It is weaker for the 10-year.
2. **The credit split I set up in advance does not reliably separate a "constructive" rally from a "stress rebound".** Rallies where credit narrowed did no better afterwards than rallies where it widened (at W=63, the "constructive" cell even had a worse next 20 sessions for the 10-year). The one consistent sign: stress rebounds (KRE up, spreads wider) were followed by KRE lagging IWM over 60 sessions in most variants. But n is 5–10 per cell, the effect shrinks to about −2% outside the GFC, and the alternative thresholds leave 1–3 cases. **The data cannot say which kind of rally this is with any confidence.** That is the honest answer to the central question.
3. **The rate-beta flip is real, but it is neither new nor specific to KRE.** KRE's 63-session sensitivity to the 10-year (controlling for SPY) is −0.062 per 1pp. That is the 4th percentile since 2012 and the 4th percentile since 2006. A negative beta was normal in 2006–2011, absent in 2013–2021, and has been the *majority* state since August 2023 (57% of days; record low −0.157 in May 2024). **KBWB's rate beta is negative too (−0.031, 4th percentile), and the two rolling betas correlate at 0.98.** So this is a bank-sector regime, not a KRE-only one. What *is* unusual is a negative beta during a Fed hiking phase: in 2015–18 it was never negative, and in 2022–23 it was negative on only 19% of days.
4. **Where 10/2026 sits:** the trailing 20-session window (10/2–10/30) falls in the pre-defined **"stress rebound"** cell. The 2-year fell 27bp and the 10-year 34bp (a bull flattener). KRE rose 1.7% and beat KBWB, while the Baa spread, HYG-vs-IEF and HY OAS all widened. The 10/14–10/29 fall in the 10-year was **entirely real yield; 10-year breakevens were unchanged** (FRED DFII10/T10YIE). That fits growth fear or safe-haven buying, not disinflation. A 10/22-type day (10-year −17bp, SPY −1.2%, KRE +2.5%) has **one precedent since 2006: 1/22/2008**. The closest mechanical analogs are the post-2023 "AOCI-relief" rallies (Dec 2023, May 2024, Jan 2025, Sep 2025, Feb 2026, Jun 2026). Five of those six had negative forward 60-session returns. Across all 10 post-2012 analogs, 7 were negative and the mean was −4.3%, against a +2.0% base. This is descriptive only, and the cases overlap with known shocks (tariffs in 2025).
5. **Cross-section (167 KRE names, June-2026 call reports):** within KRE, sensitivity to the 10-year runs through **size, CRE concentration, loan/deposit ratio and deposit costs**. It does *not* run through securities losses. Banks with bigger HTM losses relative to Tier 1 did better on *both* falling- and rising-yield days. So the earlier "bigger HTM losses → bigger rally" result is general outperformance, not specific rate relief. Thin sample: only 8 falling and 14 rising days of ≥5bp since 7/1.

Net read. The current rally looks more like a flight-to-quality rebound inside a hiking cycle than a constructive easing rally. History gives a mild lean toward KRE giving back relative performance (against KBWB, and probably against IWM) over the next ~60 sessions. It gives **no reliable edge on KRE's absolute direction.** I would not call it "constructive" or "temporary" on this evidence.

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0. Baseline re-verification (prior report not attached; re-derived from raw data)

The earlier PDF was **not attached** to this session (the uploads folder was empty), so every number below was rebuilt from primary data and compared with the figures in the brief.

Brief's number	Re-derived	Status
KRE–KBWB 63-session corr 0.719 on 10/30, 1.5th pct of 3,708 windows since Feb 2012	0.7193; 1.54th pct; 3,708 windows from 2012-02-02	✓
Low 0.632 on 9/15/2026	0.6324 on 2026-09-15	✓
8/3→10/30: KRE −6.0%, KBWB −8.4%	−6.02%, −8.43% (IWM −5.46%, SPY −0.01%)	✓
8/3→10/14: KRE −10.6%, KBWB −11.5%	−10.55%, −11.47%	✓
10/14→10/30: KRE +5.1%, KBWB +3.4%, IWM +2.9%	+5.06%, +3.43%, +2.93%	✓
Rate coefficient −0.061 per 1pp, 4th pct, median +0.070	**−0.062**; 4.2th pct since Feb 2012 (median **+0.071**); 3.7th pct since 2006 (median +0.042)	✓ (the brief's median is a since-2012 figure)
SPY beta 0.21 (1.6th pct)	**0.24** (1.3th pct, full history)	≈ (small difference, probably yield source/window alignment; I use Treasury par yields)
R^2 0.17; rates explain ~+1.6pp of the +5.0pp rally	R^2 **0.15**; −0.062 × (−25bp) = **+1.55pp**	≈
10y 4.70 → 5.19 → 4.94; 2y 4.25 → 4.82 → 4.56	identical (Treasury par curve)	✓
HY OAS 2.78 → 3.25 → 3.44	identical (FRED BAMLH0A0HYM2)	✓
10/22: 10y −17bp, 2y −16bp; KRE +2.5% with 96% of stocks up; SPY −1.2%; IWM flat; HY wider	10y −17bp, 2y −16bp; KRE +2.53%; 96.4% of the 167 up; SPY −1.18%; IWM +0.07%; HY OAS 3.27 → 3.40	✓
Fed: +25bp in Sept to 3.75–4.00%; 10/28 hold 10–2, two hike dissents	Confirmed in Fed press releases (9/16/2026 statement; 10/28/2026 statement "by a 10 – 2 vote"). Dissenter names (Cook, Hammack) are from press coverage (CNBC, J.P. Morgan AM).	✓
"Futures price a December hike"	Secondary only (CNBC 10/28: "strong possibility" of a December hike); not checked against CME data	secondary
HTM-loss/T1 → bigger rally (t ≈ −2.1 to −2.6)	Not re-run in the same form. §4 re-tests it on falling- vs rising-yield days, and the reading changes.	see §4

Data limitations that bind this study

- **HY OAS history is truncated.** The free FRED CSV for BAMLH0A0HYM2 now starts on **2023-11-03**. The FRED series page says that "starting in April 2026, this series will only include 3 years of observations" (ICE licence). ALFRED vintages returned HTTP 503 from this environment on 11/02. **HY OAS is therefore used only for the current episode.** Long-history classification uses two proxies, fixed in advance:
  - i. **Moody's Baa − 10y (FRED BAA10Y)**, from 1986. This is the primary measure.
  - ii. **HYG − IEF total-return difference**, from April 2007, as a cross-check.
  - In the 2023-11 → 2026-10 overlap, 20-day changes in HY OAS correlate +0.65 with ΔBAA10Y and −0.76 with HYG−IEF. But the *direction* agrees only 70% and 73% of the time. These are rough stand-ins, and they are the largest single weakness of sub-question 1.
- **KBWB starts on 2011-10-20.** Every "vs KBWB" statistic covers 2012–2026 only (59 of 80 W=63 windows; 188 of 255 W=20 windows).
- **KRE starts on 2006-06-22.** So the sample has one full hiking cycle (2015–18), one fast one (2022–23), and the start of the 2026 phase.
- FDIC call reports for 9/30/2026 are **not loaded** (the API returned 0 records for REPDTE=20260930 on 11/02). The cross-section uses 6/30/2026.

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1. Historical regimes: falling yields on non-overlapping windows

Method (fixed before any forward return was computed; full text in `PREREGISTERED_SPEC.md`).

- **Windows:** non-overlapping blocks of 20 and 63 sessions, anchored so the last block ends on 10/30/2026.
- **Falling yields:** Δ2y < 0 or Δ10y < 0 over the window, with the two tenors reported separately.
- **Classifications:**
  - a. KRE return > 0;
  - b. Δ(Baa−10y) > 0 = credit widening (HYG−IEF < 0 as a cross-check);
  - c. Δ(10y−2y) > 0 = steepening;
  - d. KRE − KBWB > 0.
- **Forward returns:** the next 20 and 60 sessions from the window's end close.
- **Overlap control:** for W=20 with a 60-session horizon, only every third window is used, so forward periods never overlap. The other two phases are shown in Table 1f.
- **Statistics:** the bracket shows the difference from the **unconditional baseline**, meaning all windows on the same grid. ** means |Welch t| ≥ 2 for the cell against its complement (computed only when n ≥ 10). † means n < 10: too thin to support a conclusion.
- **Pre-defined labels:** CONSTRUCTIVE = falling yields + KRE up + credit narrowing. STRESS REBOUND = falling yields + KRE up + credit widening.

How to read the tables: "n (ep.) 20d" gives windows with a forward-20 return, the number of separate episodes (runs of consecutive windows) in brackets, and then the count with KBWB data. Returns are in %. A bracket is the difference from baseline in pp.

Table 1a — W = 63 sessions, 2-year yield

Cell	n (ep.) 20d	n 60d	KRE F20	vs KBWB F20	vs IWM F20	KRE F60	vs KBWB F60	vs IWM F60
Baseline: all windows	80 / 59	80 / 59	+1.2 (55%↑)	+0.0 (53%↑)	+0.4 (51%↑)	+2.0 (59%↑)	-0.5 (46%↑)	-0.5 (46%↑)
Yields falling	37 (20) / 25	37 / 25	+0.7 [-0.5]	+0.0 [-0.0]	-0.5 [-0.9]	+0.6 [-1.4]	**-2.2 [-1.7]**	-2.4 [-1.9]
Yields not falling	43 (20) / 34	43 / 34	+1.6 [+0.4]	+0.0 [+0.0]	+1.2 [+0.8]	+3.2 [+1.2]	**+0.8 [+1.2]**	+1.1 [+1.6]
↳ KRE up	22 (15) / 17	22 / 17	+0.8 [-0.4]	-0.0 [-0.1]	-0.6 [-1.1]	+0.4 [-1.6]	**-2.4 [-1.9]**	-2.2 [-1.7]
↳ KRE down	15 (12) / 8	15 / 8	+0.6 [-0.6]	+0.1 [+0.1]†	-0.3 [-0.8]	+0.8 [-1.2]	-1.8 [-1.3]†	-2.7 [-2.2]
↳ Baa spread widening	22 (17) / 11	22 / 11	+0.3 [-0.9]	+0.4 [+0.4]	-0.9 [-1.3]	+0.7 [-1.3]	-1.6 [-1.1]	-3.1 [-2.6]
↳ Baa spread narrowing	15 (13) / 14	15 / 14	+1.4 [+0.1]	-0.2 [-0.3]	+0.1 [-0.4]	+0.3 [-1.7]	**-2.6 [-2.2]**	-1.3 [-0.9]
↳ HYG<IEF (widening)	19 (14) / 12	19 / 12	+0.5 [-0.7]	+0.7 [+0.7]	-0.2 [-0.6]	+0.7 [-1.3]	-1.8 [-1.4]	-2.5 [-2.0]
↳ HYG>IEF (narrowing)	16 (14) / 13	16 / 13	+1.1 [-0.1]	-0.6 [-0.6]	-0.8 [-1.2]	+0.7 [-1.3]	**-2.5 [-2.0]**	-1.9 [-1.4]
↳ 2s10s steepening	20 (13) / 13	20 / 13	+0.8 [-0.4]	+0.6 [+0.5]	-0.2 [-0.7]	-0.0 [-2.0]	-1.6 [-1.1]	-2.3 [-1.8]
↳ 2s10s flattening	17 (13) / 12	17 / 12	+0.7 [-0.6]	-0.5 [-0.6]	-0.8 [-1.2]	+1.2 [-0.8]	**-2.8 [-2.3]**	-2.5 [-2.0]
↳ KRE beat KBWB	12 (11) / 12	12 / 12	+1.4 [+0.1]	-0.5 [-0.5]	-0.2 [-0.6]	+0.8 [-1.2]	-2.4 [-1.9]	-1.1 [-0.6]
↳ KRE lagged KBWB	12 (8) / 12	12 / 12	+2.6 [+1.4]	+0.5 [+0.5]	-0.5 [-1.0]	+1.9 [-0.1]	-1.7 [-1.2]	-2.9 [-2.4]
↳ CONSTRUCTIVE (KRE up, Baa narrowing)	12 (11) / 11	12 / 11	+1.6 [+0.4]	-0.4 [-0.4]	-0.0 [-0.5]	+2.6 [+0.6]	-2.6 [-2.1]	+0.1 [+0.6]
↳ STRESS REBOUND (KRE up, Baa widening)	10 (10) / 6	10 / 6	-0.2 [-1.4]	+0.6 [+0.6]†	**-1.3 [-1.8]**	-2.1 [-4.2]	-2.0 [-1.5]†	**-4.9 [-4.4]**
↳ STRESS (KRE down, Baa widening)	12 (10) / 5	12 / 5	+0.7 [-0.5]	+0.1 [+0.1]†	-0.5 [-0.9]	+3.1 [+1.1]	-1.2 [-0.7]†	-1.5 [-1.0]
↳ BENIGN/OTHER (KRE down, Baa narrowing)	3 (3) / 3	3 / 3	+0.2 [-1.0]†	+0.1 [+0.1]†	+0.4 [-0.0]†	-8.6 [-10.6]†	-2.9 [-2.4]†	-7.2 [-6.7]†
↳ CONSTRUCTIVE, HYG proxy	14 (13) / 11	14 / 11	+1.4 [+0.1]	-0.4 [-0.4]	-0.9 [-1.3]	+0.6 [-1.4]	-2.5 [-2.0]	-2.3 [-1.8]
↳ STRESS REBOUND, HYG proxy	7 (6) / 6	7 / 6	-0.3 [-1.5]†	+0.7 [+0.6]†	+0.2 [-0.3]†	+0.1 [-1.9]†	-2.1 [-1.6]†	-1.8 [-1.3]†

Table 1b — W = 63 sessions, 10-year yield

Cell	n (ep.) 20d	n 60d	KRE F20	vs KBWB F20	vs IWM F20	KRE F60	vs KBWB F60	vs IWM F60
Baseline: all windows	80 / 59	80 / 59	+1.2 (55%↑)	+0.0 (53%↑)	+0.4 (51%↑)	+2.0 (59%↑)	-0.5 (46%↑)	-0.5 (46%↑)
Yields falling	39 (20) / 28	39 / 28	-0.2 [-1.4]	-0.2 [-0.2]	**-0.7 [-1.1]**	+0.7 [-1.4]	**-1.7 [-1.3]**	-1.8 [-1.3]
Yields not falling	41 (20) / 31	41 / 31	+2.6 [+1.4]	+0.2 [+0.2]	**+1.5 [+1.0]**	+3.3 [+1.3]	**+0.7 [+1.1]**	+0.8 [+1.3]
↳ KRE up	22 (15) / 18	22 / 18	**-0.9 [-2.2]**	-0.4 [-0.4]	**-1.1 [-1.6]**	+0.2 [-1.8]	-1.9 [-1.4]	-2.0 [-1.5]
↳ KRE down	17 (12) / 10	17 / 10	+0.7 [-0.5]	+0.2 [+0.1]	-0.1 [-0.5]	+1.2 [-0.8]	-1.4 [-0.9]	-1.6 [-1.1]
↳ Baa spread widening	23 (17) / 13	23 / 13	+0.5 [-0.7]	+0.0 [-0.0]	-0.5 [-1.0]	+1.4 [-0.6]	-1.7 [-1.2]	-2.1 [-1.6]
↳ Baa spread narrowing	16 (13) / 15	16 / 15	-1.2 [-2.4]	-0.4 [-0.4]	-0.8 [-1.3]	-0.4 [-2.4]	-1.8 [-1.3]	-1.4 [-0.9]
↳ HYG<IEF (widening)	22 (15) / 15	22 / 15	+0.8 [-0.4]	+0.7 [+0.6]	+0.1 [-0.4]	+1.4 [-0.6]	-1.3 [-0.8]	-1.5 [-1.0]
↳ HYG>IEF (narrowing)	15 (12) / 13	15 / 13	**-1.7 [-2.9]**	-1.1 [-1.1]	**-1.7 [-2.1]**	-0.1 [-2.1]	-2.3 [-1.8]	-1.9 [-1.4]
↳ 2s10s steepening	14 (10) / 9	14 / 9	-1.0 [-2.2]	+0.1 [+0.0]†	-1.1 [-1.6]	-1.4 [-3.4]	-1.6 [-1.1]†	-3.0 [-2.5]
↳ 2s10s flattening	25 (15) / 19	25 / 19	+0.2 [-1.0]	-0.3 [-0.3]	-0.4 [-0.8]	+1.8 [-0.2]	-1.8 [-1.3]	-1.2 [-0.7]
↳ KRE beat KBWB	10 (8) / 10	10 / 10	+0.7 [-0.6]	-0.9 [-0.9]	-0.9 [-1.3]	-0.2 [-2.2]	-2.5 [-2.0]	-2.0 [-1.5]
↳ KRE lagged KBWB	17 (11) / 17	17 / 17	+0.4 [-0.8]	+0.3 [+0.2]	-0.7 [-1.1]	+1.0 [-1.0]	-1.1 [-0.6]	-2.2 [-1.7]
↳ CONSTRUCTIVE (KRE up, Baa narrowing)	12 (10) / 11	12 / 11	**-2.0 [-3.2]**	-0.5 [-0.6]	-1.6 [-2.0]	+0.4 [-1.6]	-1.6 [-1.1]	-1.0 [-0.5]

↳ STRESS REBOUND (KRE up, Baa widening)	10 (10) / 7	10 / 7	+0.3 [-0.9]	-0.1 [-0.1]†	-0.5 [-1.0]	-0.1 [-2.1]	-2.4 [-1.9]†	-3.2 [-2.7]
↳ STRESS (KRE down, Baa widening)	13 (10) / 6	13 / 6	+0.6 [-0.6]	+0.2 [+0.1]†	-0.5 [-0.9]	+2.5 [+0.5]	-0.8 [-0.3]†	-1.3 [-0.8]
↳ BENIGN/OTHER (KRE down, Baa narrowing)	4 (4) / 4	4 / 4	+1.2 [-0.0]†	+0.2 [+0.1]†	+1.4 [+1.0]†	-3.0 [-5.0]†	-2.3 [-1.8]†	-2.7 [-2.2]†
↳ CONSTRUCTIVE, HYG proxy	13 (11) / 11	13 / 11	**-1.8 [-3.1]**	-1.0 [-1.1]	**-1.9 [-2.4]**	-0.3 [-2.3]	-2.3 [-1.8]	-2.3 [-1.8]
↳ STRESS REBOUND, HYG proxy	8 (7) / 7	8 / 7	+0.3 [-0.9]†	+0.7 [+0.6]†	+0.3 [-0.1]†	+1.1 [-0.9]†	-1.4 [-0.9]†	-1.3 [-0.8]†

Table 1c — W = 20 sessions, 2-year yield (F60 uses every third window)

Cell	n (ep.) 20d	n 60d	KRE F20	vs KBWB F20	vs IWM F20	KRE F60	vs KBWB F60	vs IWM F60
Baseline: all windows	255 / 188	85 / 62	+0.7 (53%↑)	-0.2 (45%↑)	-0.2 (46%↑)	+2.0 (56%↑)	-0.5 (40%↑)	-0.5 (47%↑)
Yields falling	119 (60) / 77	39 / 25	+1.1 [+0.4]	-0.3 [-0.1]	-0.3 [-0.1]	+0.3 [-1.7]	**-1.9 [-1.5]**	**-2.7 [-2.2]**
Yields not falling	136 (60) / 111	46 / 37	+0.3 [-0.4]	-0.1 [+0.1]	-0.1 [+0.1]	+3.5 [+1.5]	**+0.5 [+1.0]**	**+1.4 [+1.9]**
↳ KRE up	57 (41) / 37	18 / 11	+1.2 [+0.5]	-0.1 [+0.1]	-0.2 [+0.0]	-0.3 [-2.3]	**-3.2 [-2.8]**	-2.1 [-1.6]
↳ KRE down	62 (43) / 40	21 / 14	+1.0 [+0.4]	-0.4 [-0.3]	-0.4 [-0.3]	+0.8 [-1.2]	-0.9 [-0.4]	-3.2 [-2.7]
↳ Baa spread widening	68 (44) / 41	22 / 14	+1.8 [+1.1]	-0.1 [+0.0]	-0.1 [+0.1]	-0.5 [-2.5]	-1.1 [-0.6]	-3.4 [-2.9]
↳ Baa spread narrowing	51 (42) / 36	17 / 11	+0.2 [-0.5]	-0.4 [-0.3]	-0.6 [-0.5]	+1.3 [-0.7]	**-3.0 [-2.5]**	-1.8 [-1.3]
↳ HYG<IEF (widening)	67 (47) / 43	25 / 15	+1.2 [+0.5]	-0.2 [+0.0]	-0.0 [+0.2]	-2.1 [-4.1]	-1.4 [-0.9]	**-4.3 [-3.8]**
↳ HYG>IEF (narrowing)	46 (38) / 34	13 / 10	+1.1 [+0.4]	-0.4 [-0.2]	-0.5 [-0.3]	+4.6 [+2.6]	**-2.7 [-2.2]**	+0.8 [+1.3]
↳ 2s10s steepening	51 (35) / 30	14 / 7	+1.1 [+0.4]	-0.6 [-0.4]	+0.2 [+0.3]	+3.6 [+1.5]	-2.2 [-1.7]†	+1.1 [+1.6]
↳ 2s10s flattening	68 (51) / 47	25 / 18	+1.1 [+0.5]	-0.0 [+0.1]	-0.7 [-0.5]	-1.5 [-3.6]	-1.8 [-1.4]	**-4.9 [-4.3]**
↳ KRE beat KBWB	34 (27) / 34	12 / 12	**+3.5 [+2.8]**	+0.0 [+0.2]	**+1.5 [+1.7]**	+3.1 [+1.1]	-2.6 [-2.2]	-0.9 [-0.4]
↳ KRE lagged KBWB	42 (32) / 42	13 / 13	+0.6 [-0.1]	-0.5 [-0.4]	-1.1 [-0.9]	+1.5 [-0.5]	-1.3 [-0.8]	-1.0 [-0.4]
↳ CONSTRUCTIVE (KRE up, Baa narrowing)	34 (32) / 25	13 / 9	+0.6 [-0.1]	-0.3 [-0.1]	-0.7 [-0.5]	+1.8 [-0.3]	-3.2 [-2.7]†	-1.8 [-1.3]
↳ STRESS REBOUND (KRE up, Baa widening)	23 (18) / 12	5 / 2	+2.1 [+1.4]	+0.4 [+0.6]	+0.6 [+0.8]	-5.5 [-7.5]†	-3.6 [-3.1]†	-2.8 [-2.3]†
↳ STRESS (KRE down, Baa widening)	45 (34) / 29	17 / 12	+1.7 [+1.0]	-0.4 [-0.2]	-0.4 [-0.2]	+1.0 [-1.1]	-0.7 [-0.2]	-3.6 [-3.1]
↳ BENIGN/OTHER (KRE down, Baa narrowing)	17 (16) / 11	4 / 2	-0.7 [-1.3]	-0.7 [-0.6]	-0.5 [-0.4]	-0.1 [-2.1]†	-2.0 [-1.5]†	-1.7 [-1.2]†
↳ CONSTRUCTIVE, HYG proxy	30 (27) / 23	10 / 8	+1.7 [+1.0]	-0.4 [-0.2]	-0.4 [-0.2]	+6.0 [+4.0]	-2.9 [-2.4]†	+1.9 [+2.4]
↳ STRESS REBOUND, HYG proxy	22 (19) / 14	7 / 3	+0.6 [-0.0]	+0.4 [+0.6]	+0.5 [+0.6]	-9.8 [-11.8]†	-4.3 [-3.8]†	-7.1 [-6.6]†

Table 1d — W = 20 sessions, 10-year yield

Cell	n (ep.) 20d	n 60d	KRE F20	vs KBWB F20	vs IWM F20	KRE F60	vs KBWB F60	vs IWM F60
Baseline: all windows	255 / 188	85 / 62	+0.7 (53%↑)	-0.2 (45%↑)	-0.2 (46%↑)	+2.0 (56%↑)	-0.5 (40%↑)	-0.5 (47%↑)
Yields falling	128 (67) / 91	43 / 32	+0.9 [+0.2]	+0.0 [+0.2]	-0.3 [-0.1]	-0.2 [-2.2]	-0.9 [-0.4]	-2.4 [-1.9]
Yields not falling	127 (67) / 97	42 / 30	+0.5 [-0.2]	-0.3 [-0.2]	-0.0 [+0.1]	+4.3 [+2.3]	-0.1 [+0.4]	+1.4 [+2.0]
↳ KRE up	59 (47) / 42	18 / 12	+0.3 [-0.4]	+0.1 [+0.2]	-0.3 [-0.1]	**-3.2 [-5.2]**	**-3.5 [-3.0]**	-3.1 [-2.6]
↳ KRE down	69 (49) / 49	25 / 20	+1.4 [+0.7]	-0.0 [+0.1]	-0.3 [-0.1]	+1.9 [-0.1]	+0.7 [+1.2]	-1.9 [-1.4]
↳ Baa spread widening	70 (48) / 48	25 / 18	+1.7 [+1.1]	+0.2 [+0.4]	+0.0 [+0.2]	-0.9 [-2.9]	-0.7 [-0.2]	-3.6 [-3.1]
↳ Baa spread narrowing	58 (49) / 43	18 / 14	-0.1 [-0.8]	-0.2 [-0.1]	-0.7 [-0.5]	+0.8 [-1.3]	-1.1 [-0.6]	-0.7 [-0.2]
↳ HYG<IEF (widening)	80 (57) / 58	29 / 20	+0.9 [+0.2]	-0.0 [+0.1]	-0.1 [+0.1]	-1.0 [-3.0]	-0.4 [+0.1]	**-3.7 [-3.2]**
↳ HYG>IEF (narrowing)	43 (39) / 33	13 / 12	+0.9 [+0.2]	+0.1 [+0.3]	-0.4 [-0.2]	+1.3 [-0.7]	-1.7 [-1.2]	+0.7 [+1.3]
↳ 2s10s steepening	30 (21) / 17	8 / 4	+0.9 [+0.2]	-0.3 [-0.1]	-0.4 [-0.2]	-1.7 [-3.8]†	-3.0 [-2.5]†	-0.4 [+0.2]†
↳ 2s10s flattening	98 (64) / 74	35 / 28	+0.9 [+0.2]	+0.1 [+0.2]	-0.3 [-0.1]	+0.1 [-1.9]	-0.6 [-0.1]	**-2.9 [-2.4]**
↳ KRE beat KBWB	37 (29) / 37	13 / 13	+2.2 [+1.6]	-0.0 [+0.2]	+0.7 [+0.9]	+0.5 [-1.6]	-2.6 [-2.1]	-2.3 [-1.7]
↳ KRE lagged KBWB	53 (36) / 53	19 / 19	+0.2 [-0.5]	-0.0 [+0.1]	-0.5 [-0.3]	+2.8 [+0.8]	+0.3 [+0.8]	+0.6 [+1.2]
↳ CONSTRUCTIVE (KRE up, Baa narrowing)	37 (35) / 28	13 / 9	-0.3 [-1.0]	-0.1 [+0.0]	-0.8 [-0.7]	-0.6 [-2.6]	-3.3 [-2.8]†	-2.3 [-1.8]
↳ STRESS REBOUND (KRE up, Baa widening)	22 (17) / 14	5 / 3	+1.3 [+0.6]	+0.5 [+0.7]	+0.6 [+0.8]	-9.9 [-11.9]†	-4.0 [-3.6]†	-5.3 [-4.8]†
↳ STRESS (KRE down, Baa widening)	48 (38) / 34	20 / 15	+1.9 [+1.2]	+0.1 [+0.3]	-0.2 [-0.1]	+1.4 [-0.7]	-0.0 [+0.5]	-3.2 [-2.7]
↳ BENIGN/OTHER (KRE down, Baa narrowing)	21 (19) / 15	5 / 5	+0.2 [-0.5]	-0.3 [-0.2]	-0.5 [-0.4]	+4.2 [+2.2]†	+2.8 [+3.3]†	+3.3 [+3.8]†
↳ CONSTRUCTIVE, HYG proxy	28 (27) / 23	9 / 8	+1.0 [+0.3]	+0.1 [+0.3]	-0.2 [-0.0]	+1.9 [-0.1]†	-3.0 [-2.5]†	+1.0 [+1.5]†
↳ STRESS REBOUND, HYG proxy	27 (24) / 19	8 / 4	-0.5 [-1.1]	+0.0 [+0.2]	-0.2 [-0.0]	-9.7 [-11.7]†	-4.5 [-4.0]†	-7.3 [-6.8]†

Table 1e — Robustness of the key cells: alternative thresholds, ex-GFC (60-session horizon)

Alternative thresholds: falling yields means Δy ≤ −10bp (W=20) or ≤ −20bp (W=63), plus dead-bands on every dimension. Ex-GFC drops windows that touch 2007-07 → 2009-06.

W	Tenor	Cell	F60 KRE: primary	alt thresholds	ex-GFC	F60 vs KBWB: primary	alt
20	2y	Yields falling	+0.3 (n=39)	-4.2 (n=21)	+2.2 (n=32)	-1.9 (n=25)	-2.3 (n=12)
20	2y	KRE up	-0.3 (n=18)	-3.7 (n=6)	+3.5 (n=15)	-3.2 (n=11)	-7.0 (n=2)
20	2y	CONSTRUCTIVE	+1.8 (n=13)	+3.4 (n=3)	+3.2 (n=12)	-3.2 (n=9)	-7.0 (n=2)
20	2y	STRESS REBOUND	-5.5 (n=5)	-20.3 (n=1)	+5.0 (n=3)	-3.6 (n=2)	—
20	2y	STRESS REBOUND (HYG)	-9.8 (n=7)	-10.7 (n=3)	-5.2 (n=5)	-4.3 (n=3)	-12.6 (n=1)
20	10y	Yields falling	-0.2 (n=43)	-0.7 (n=28)	+1.6 (n=38)	-0.9 (n=32)	-0.9 (n=18)
20	10y	KRE up	-3.2 (n=18)	-1.5 (n=7)	+0.1 (n=15)	-3.5 (n=12)	-4.9 (n=4)
20	10y	CONSTRUCTIVE	-0.6 (n=13)	+3.4 (n=3)	+0.7 (n=12)	-3.3 (n=9)	-7.0 (n=2)
20	10y	STRESS REBOUND	-9.9 (n=5)	-20.3 (n=1)	-2.3 (n=3)	-4.0 (n=3)	—
20	10y	STRESS REBOUND (HYG)	-9.7 (n=8)	-10.7 (n=3)	-5.8 (n=6)	-4.5 (n=4)	-12.6 (n=1)
63	2y	Yields falling	+0.6 (n=37)	+0.6 (n=21)	+2.2 (n=32)	-2.2 (n=25)	-2.9 (n=12)
63	2y	KRE up	+0.4 (n=22)	-4.0 (n=5)	+3.0 (n=20)	-2.4 (n=17)	-7.1 (n=3)
63	2y	CONSTRUCTIVE	+2.6 (n=12)	-4.2 (n=2)	+2.6 (n=12)	-2.6 (n=11)	-8.0 (n=2)
63	2y	STRESS REBOUND	-2.1 (n=10)	-3.8 (n=3)	+3.5 (n=8)	-2.0 (n=6)	-5.2 (n=1)
63	2y	STRESS REBOUND (HYG)	+0.1 (n=7)	-11.8 (n=2)	+4.3 (n=6)	-2.1 (n=6)	-5.2 (n=1)
63	10y	Yields falling	+0.7 (n=39)	+2.1 (n=24)	+1.5 (n=35)	-1.7 (n=28)	-2.2 (n=16)
63	10y	KRE up	+0.2 (n=22)	+2.7 (n=3)	+1.4 (n=21)	-1.9 (n=18)	-9.4 (n=2)
63	10y	CONSTRUCTIVE	+0.4 (n=12)	-5.8 (n=1)	+0.4 (n=12)	-1.6 (n=11)	-13.6 (n=1)
63	10y	STRESS REBOUND	-0.1 (n=10)	+7.0 (n=2)	+2.7 (n=9)	-2.4 (n=7)	-5.2 (n=1)
63	10y	STRESS REBOUND (HYG)	+1.1 (n=8)	+1.8 (n=1)	+4.9 (n=7)	-1.4 (n=7)	-5.2 (n=1)

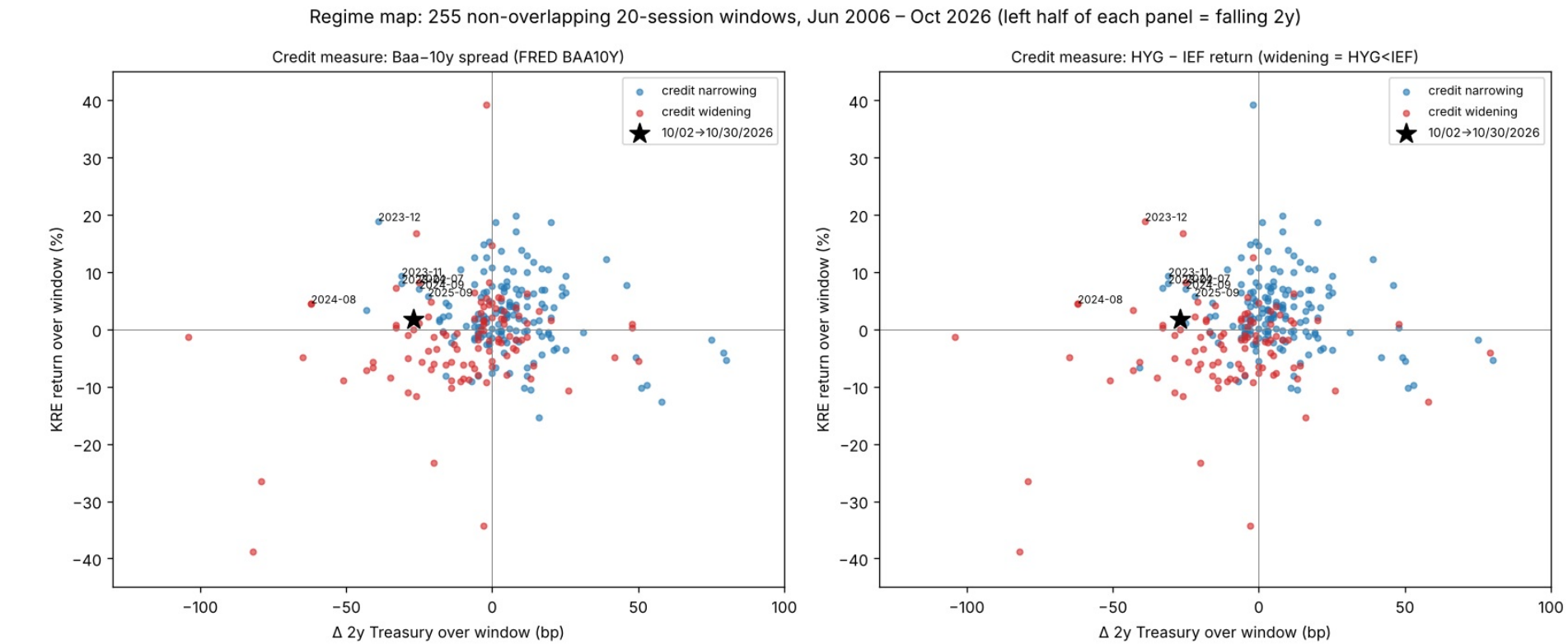
Table 1f — W = 20, 60-session horizon: sensitivity to the grid phase

Tenor	Cell	phase 0 (primary)	phase 1	phase 2
y2	ALL	+2.0 (n=85, 56%↑)	+1.8 (n=84, 58%↑)	+2.0 (n=84, 63%↑)
y2	falling	+0.3 (n=39, 51%↑)	+0.9 (n=43, 56%↑)	+2.2 (n=37, 62%↑)
y2	falling&KRE up&Baa widening	-5.5 (n=5, 40%↑)	+0.9 (n=11, 55%↑)	+3.8 (n=7, 71%↑)
y2	falling&KRE up&Baa narrowing	+1.8 (n=13, 46%↑)	+0.6 (n=11, 45%↑)	+3.3 (n=10, 60%↑)
y2	falling&KRE up&HYG<IEF	-9.8 (n=7, 0%↑)	+1.1 (n=9, 44%↑)	+10.5 (n=6, 100%↑)
y2	falling&KRE up&HYG>IEF	+6.0 (n=10, 70%↑)	+0.9 (n=11, 55%↑)	-0.9 (n=9, 33%↑)
y10	ALL	+2.0 (n=85, 56%↑)	+1.8 (n=84, 58%↑)	+2.0 (n=84, 63%↑)
y10	falling	-0.2 (n=43, 51%↑)	+1.4 (n=45, 58%↑)	+3.0 (n=40, 62%↑)
y10	falling&KRE up&Baa widening	-9.9 (n=5, 20%↑)	+1.1 (n=9, 44%↑)	+5.0 (n=8, 88%↑)
y10	falling&KRE up&Baa narrowing	-0.6 (n=13, 46%↑)	+0.4 (n=11, 45%↑)	+1.9 (n=13, 54%↑)
y10	falling&KRE up&HYG<IEF	-9.7 (n=8, 0%↑)	+0.1 (n=10, 40%↑)	+7.6 (n=9, 100%↑)
y10	falling&KRE up&HYG>IEF	+1.9 (n=9, 67%↑)	+1.1 (n=9, 44%↑)	-0.8 (n=10, 30%↑)

Table 1g — Alternative-threshold version of Table 1a (shows how thin the cells get)

Cell	n (ep.) 20d	n 60d	KRE F20	vs KBWB F20	vs IWM F20	KRE F60	vs KBWB F60	vs IWM F60
Baseline: all windows	80 / 59	80 / 59	+1.2 (55%↑)	+0.0 (53%↑)	+0.4 (51%↑)	+2.0 (59%↑)	-0.5 (46%↑)	-0.5 (46%↑)
Yields falling	21 (13) / 12	21 / 12	-0.6 [-1.8]	+0.3 [+0.2]	-1.0 [-1.5]	+0.6 [-1.4]	**-2.9 [-2.4]**	**-3.3 [-2.8]**
Yields not falling	59 (13) / 47	59 / 47	+1.8 [+0.6]	-0.0 [-0.1]	+1.0 [+0.5]	+2.5 [+0.5]	**+0.1 [+0.6]**	**+0.5 [+1.0]**
↳ KRE up	5 (5) / 3	5 / 3	-5.4 [-6.6]†	-3.0 [-3.1]†	-2.6 [-3.1]†	-4.0 [-6.0]†	-7.1 [-6.6]†	-4.6 [-4.1]†
↳ KRE down	7 (6) / 3	7 / 3	+0.4 [-0.9]†	-0.4 [-0.4]†	-2.2 [-2.6]†	+6.9 [+4.9]†	-1.5 [-1.0]†	-2.6 [-2.1]†
↳ Baa spread widening	12 (10) / 5	12 / 5	-0.8 [-2.0]	+0.6 [+0.5]†	-0.9 [-1.3]	+1.0 [-1.0]	-2.1 [-1.6]†	-3.5 [-3.0]
↳ Baa spread narrowing	4 (4) / 4	4 / 4	+1.3 [+0.1]†	-0.4 [-0.4]†	-0.7 [-1.1]†	-6.6 [-8.6]†	-6.1 [-5.6]†	-6.5 [-6.0]†
↳ HYG<IEF (widening)	15 (11) / 9	15 / 9	-0.9 [-2.1]	+0.9 [+0.9]†	-0.8 [-1.3]	+0.1 [-1.8]	-1.6 [-1.1]†	-3.6 [-3.1]
↳ HYG>IEF (narrowing)	4 (4) / 3	4 / 3	+0.4 [-0.8]†	-1.7 [-1.8]†	-1.7 [-2.2]†	+3.5 [+1.5]†	-6.9 [-6.4]†	-1.6 [-1.1]†
↳ 2s10s steepening	8 (7) / 4	8 / 4	+2.1 [+0.9]†	+0.5 [+0.4]†	+0.4 [-0.0]†	+1.9 [-0.1]†	-2.3 [-1.8]†	-2.7 [-2.2]†
↳ 2s10s flattening	6 (5) / 2	6 / 2	-2.4 [-3.6]†	+4.3 [+4.2]†	-1.8 [-2.2]†	-2.5 [-4.5]†	-2.2 [-1.7]†	-6.9 [-6.4]†
↳ KRE beat KBWB	1 (1) / 1	1 / 1	-1.2 [-2.4]†	-1.0 [-1.0]†	-0.6 [-1.0]†	+1.8 [-0.2]†	-5.2 [-4.7]†	+1.7 [+2.2]†
↳ KRE lagged KBWB	7 (5) / 7	7 / 7	+2.7 [+1.5]†	+1.5 [+1.5]†	-0.4 [-0.9]†	+4.2 [+2.1]†	-0.6 [-0.1]†	-3.1 [-2.6]†
↳ CONSTRUCTIVE (KRE up, Baa narrowing)	2 (2) / 2	2 / 2	-7.2 [-8.4]†	-4.1 [-4.1]†	-6.1 [-6.6]†	-4.2 [-6.2]†	-8.0 [-7.5]†	-4.2 [-3.7]†
↳ STRESS REBOUND (KRE up,	3 (3) / 1	3 / 1	-4.2 [-5.4]†	-1.0 [-1.0]†	-0.3 [-0.8]†	-3.8 [-5.8]†	-5.2 [-4.7]†	-4.8 [-4.3]†

↳ STRESS REBOUND (KRE up, Baa widening)	5 (5) / 1	5 / 1	-4.2 [-5.4]†	-1.0 [-1.0]†	-0.3 [-0.8]†	-5.8 [-5.8]†	-5.2 [-4.7]†	-4.8 [-4.3]†
↳ STRESS (KRE down, Baa widening)	6 (5) / 3	6 / 3	+1.9 [+0.7]†	-0.4 [-0.4]†	-1.4 [-1.8]†	+6.7 [+4.7]†	-1.5 [-1.0]†	-2.9 [-2.4]†
↳ BENIGN/OTHER (KRE down, Baa narrowing)	0 (0) / 0	0 / 0	— [—]†	— [—]†	— [—]†	— [—]†	— [—]†	— [—]†
↳ CONSTRUCTIVE, HYG proxy	3 (3) / 2	3 / 2	-5.1 [-6.3]†	-4.1 [-4.1]†	-5.0 [-5.4]†	+1.2 [-0.8]†	-8.0 [-7.5]†	-3.9 [-3.4]†
↳ STRESS REBOUND, HYG proxy	2 (2) / 1	2 / 1	-5.8 [-7.0]†	-1.0 [-1.0]†	+0.8 [+0.4]†	-11.8 [-13.8]†	-5.2 [-4.7]†	-5.5 [-5.1]†



Regime map

What sub-question 1 shows

- **Unconditional base rates.** KRE's next-60-session return averaged **+2.0%** (positive 56–59% of the time) on both grids. KRE − KBWB averaged **−0.5%** (positive 40–46%), reflecting KRE's long-run lag. Every cell has to be compared with these numbers, not with zero.
- **Falling yields alone barely change the absolute outlook.** Forward-60 KRE after falling-yield windows was −0.2% to +0.7%, against the +2.0% base. That is 1.4–2.2pp worse, but no grid or tenor reaches |t| = 2.
- **Relative to KBWB, the effect is consistent.**

Window, tenor	KRE − KBWB, next 60 sessions	Welch t
W63, 2y	−2.2%	−3.0
W63, 10y	−1.7%	−2.4
W20, 2y	−1.9%	−2.3
W20, 10y	−0.9%	−0.7

  - The alternative thresholds give the same sign (−2.3% to −2.9% for the 2-year).
  - Dropping windows whose forward period overlaps COVID (Feb–Apr 2020) or SVB (Mar–May 2023) *strengthens* it (W63, 2y: −2.5% vs +0.8% for the other windows, t −3.1).
  - The median is also negative (−1.7%).
  - The biggest contributors are the windows ending Jan 2024 (−13.6pp) and Apr 2025 (−8.9pp). Both are rallies driven by falling yields that later reversed.
- **KRE rallies during falling-yield windows are the cases that later lag KBWB.** "Falling & KRE up" gave forward-60 KRE − KBWB of −2.4% (W63, 2y, t −2.3), −3.2% (W20, 2y, t −2.6) and −3.5% (W20, 10y, t −2.9). The alternative thresholds give the same sign with 2–4 cases.
- **The credit split is weak.**

  - At W=63, “constructive” and “stress rebound” have similar forward-60 KRE returns (2y: +2.6% vs −2.1%; 10y: +0.4% vs −0.1%; n = 10–12 each).
  - The 10-year “constructive” cell had a *worse* next 20 sessions (−2.0%, t −2.2).
  - At W=20, stress rebounds look bad over 60 sessions (−5.5% to −9.9%), but only for grid phase 0. Phases 1 and 2 give +0.9% to +10.5% (Table 1f). This is not robust.
  - The one consistent sign is that stress rebounds were followed by KRE lagging IWM. That holds in 13 of the 16 primary and ex-GFC variants (Baa and HYG proxies × W × tenor), with W63/2y at −4.9% (t −2.2, n=10). Outside the GFC the effect ranges from about 0 to −7%, with n = 3–9.

- **Curve.** Bull flattening during falling yields was followed by KRE lagging IWM over 60 sessions (W20: −4.9% for the 2y and −2.9% for the 10y, both |t| > 2). Steepening cells are small (n = 8–14). This is suggestive at best.
- **Multiple testing.** 394 primary cell-vs-complement tests produced 34 with |t| ≥ 2, against about 18 expected by chance. The tests are correlated (nested cells), so treat any single starred cell with caution. The KRE-vs-KBWB pattern is the only *cluster* of results that hangs together across specifications.

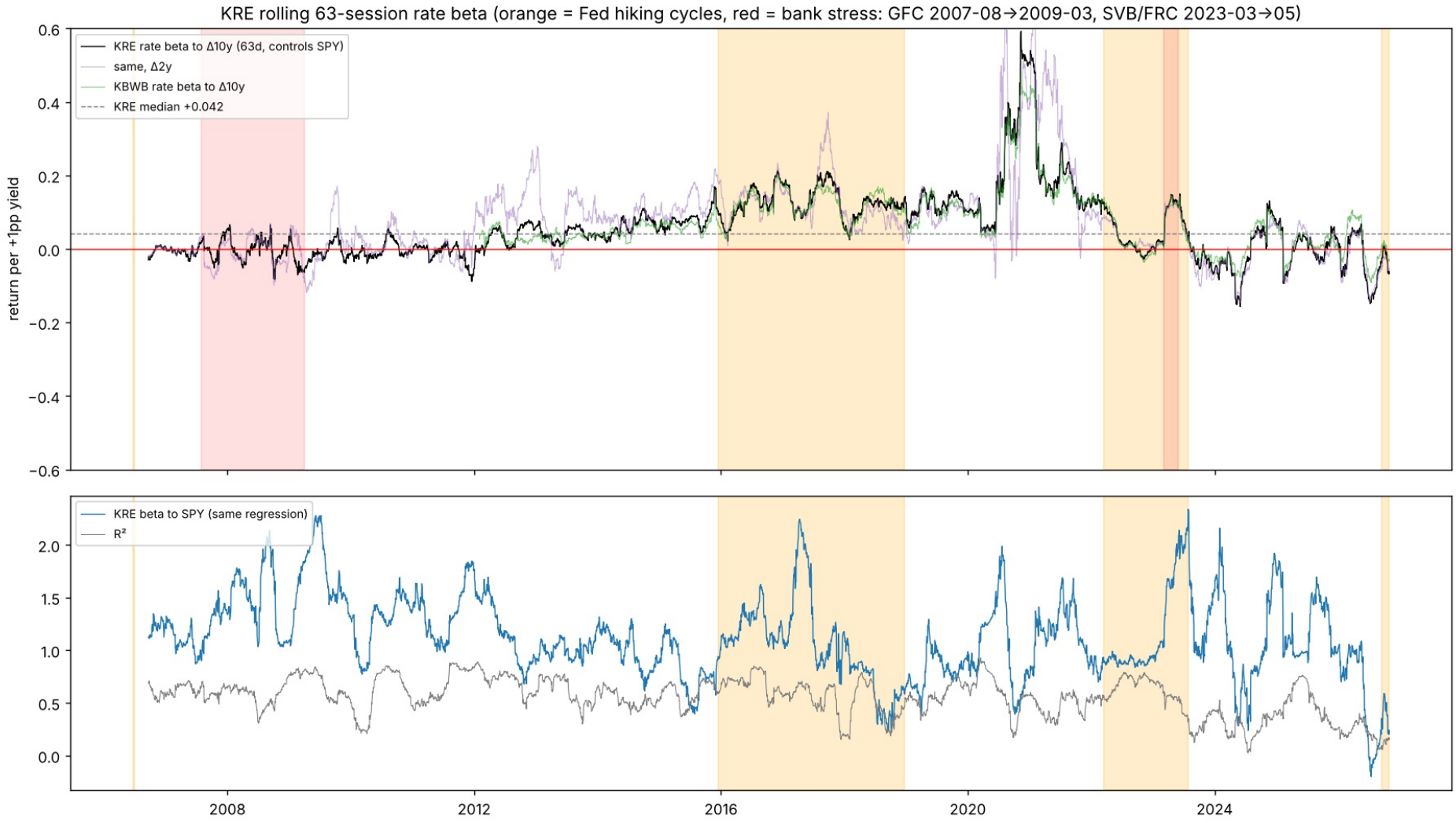
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2. The rate-beta flip

The model is a rolling 63-session OLS of r_KRE on r_SPY and the daily change in the 10-year Treasury par yield (in pp), back to 2006, with a 2-year variant. Hiking-cycle dates come from FRED DFEDTAR/DFEDTARU (effective dates):

- 2015-12-16 → 2018-12-20;
- 2022-03-17 → 2023-07-27;
- the hike announced on 2026-09-16 (effective 9/17).

Bank-stress periods are marked as GFC (2007-08 → 2009-03) and SVB/FRC (2023-03 → 2023-05).



Rolling rate beta

Table 2a — KRE rate beta by year (with KBWB for comparison)

Year	KRE rate beta: median	min	share of days <0	KBWB rate beta median	KRE SPY beta median	Fed / stress context
2006	+0.003	-0.030	46%	—	1.23	end of 2004–06 hikes (Jun)
2007	-0.005	-0.052	66%	—	1.14	cuts from Sep; GFC begins
2008	-0.003	-0.083	53%	—	1.56	GFC; ZIRP Dec
2009	-0.016	-0.070	73%	—	1.63	ZIRP
2010	-0.011	-0.040	71%	—	1.32	
2011	-0.004	-0.088	55%	—	1.35	
2012	+0.037	-0.053	14%	+0.027	1.16	
2013	+0.039	+0.007	0%	+0.034	1.00	
2014	+0.058	+0.017	0%	+0.050	1.01	
2015	+0.077	+0.038	0%	+0.066	0.78	first hike Dec 16
2016	+0.122	+0.021	0%	+0.121	1.17	hiking (1)
2017	+0.156	+0.058	0%	+0.142	1.13	hiking (3)
2018	+0.119	+0.025	0%	+0.119	0.65	hiking (4)
2019	+0.122	+0.057	0%	+0.110	0.80	3 cuts
2020	+0.162	+0.043	0%	+0.147	0.97	COVID, ZIRP
2021	+0.182	+0.103	0%	+0.164	1.20	
2022	+0.020	-0.028	25%	+0.017	0.89	hiking from Mar 16
2023	+0.019	-0.069	37%	+0.013	1.39	hiking to Jul 26; SVB/FRC Mar–May
2024	-0.027	-0.157	58%	-0.005	0.94	cuts from Sep
2025	+0.012	-0.094	40%	+0.015	1.13	cuts Sep–Dec
2026	-0.035	-0.148	63%	-0.002	0.65	HIKE Sep 16; hold Oct 28

Table 2b — Episodes of negative rate beta (b < 0; runs separated by fewer than 63 sessions are merged)

Negative-beta episode (b<0; gaps <63 sessions merged)	Sessions <0	Min beta (date)	Overlaps hiking cycle	Overlaps bank stress	KRE over span	KRE − SPY	Δ10y	Δ2y
2006-09-21 → 2010-07-13	640 of 958	-0.083 (2008-10-09)	—	GFC	-41.9%	-31.9pp	-150bp	-404bp
2010-10-13 → 2012-03-05	224 of 351	-0.088 (2011-12-19)	—	—	+14.7%	-4.3pp	-46bp	-6bp
2012-07-24 → 2012-08-02	5 of 8	-0.003 (2012-07-25)	—	—	-0.5%	-2.5pp	+7bp	+2bp
2022-09-22 → 2023-01-17	65 of 80	-0.028 (2022-11-09)	2022–23 cycle	—	-1.0%	-7.7pp	-17bp	+7bp
2023-08-15 → 2024-07-31	237 of 242	-0.157 (2024-05-31)	—	—	+33.0%	+6.9pp	-12bp	-63bp
2025-02-10 → 2026-02-26	130 of 263	-0.094 (2025-02-12)	—	—	+13.1%	-2.2pp	-49bp	-86bp
2026-05-29 → 2026-10-30	101 of 108	-0.148 (2026-07-09)	2026 hike	—	+4.7%	+4.2pp	+49bp	+58bp

Table 2c — Does a negative rate beta precede weakness? (non-overlapping 63-session grid; condition measured at the grid date)

Condition at grid date	Outcome	n	Mean	Hit rate	Baseline mean (n)	Baseline hit	Diff	t vs rest
b<0	f20_kre	20	-0.4%	40%	+1.2% (80)	55%	-1.7pp	-1.33
b<0	f60_kre	20	-0.9%	45%	+2.0% (80)	59%	-2.9pp	-1.07
b<0	f20_vs_kbwb	7	-1.8%	14%	+0.0% (59)	53%	-1.9pp	— †
b<0	f60_vs_kbwb	7	-3.0%	43%	-0.5% (59)	46%	-2.5pp	— †
b<0	f20_vs_iwm	20	-1.4%	30%	+0.4% (80)	51%	-1.9pp	-2.15
b<0	f60_vs_iwm	20	-3.6%	30%	-0.5% (80)	46%	-3.1pp	-1.91
b<0	f60_maxdd	20	-14.3%	0%	-13.2% (80)	0%	-1.1pp	-0.66
b<-0.03	f20_kre	8	-4.3%	12%	+1.2% (80)	55%	-5.5pp	— †
b<-0.03	f60_kre	8	-8.6%	25%	+2.0% (80)	59%	-10.6pp	— †
b<-0.03	f20_vs_kbwb	4	-2.2%	25%	+0.0% (59)	53%	-2.2pp	— †
b<-0.03	f60_vs_kbwb	4	-0.3%	75%	-0.5% (59)	46%	+0.2pp	— †
b<-0.03	f20_vs_iwm	8	-3.9%	25%	+0.4% (80)	51%	-4.3pp	— †
b<-0.03	f60_vs_iwm	8	-7.6%	12%	-0.5% (80)	46%	-7.1pp	— †
b<-0.03	f60_maxdd	8	-20.3%	0%	-13.2% (80)	0%	-7.1pp	— †

Onsets of negative-beta regimes (first day with b < 0 after ≥63 sessions with b ≥ 0). Descriptive only. KRE − KBWB over the next 60 sessions:

Onset	Result
2012-07	−8.6%
2022-09	−4.6%
2023-08	−1.8%

2024-11 (1-day blip)	−4.5%
2025-02	−1.1%
2026-05	−3.9%
2010-10	no KBWB data yet

That is negative in 6 of 6 cases with data, against a 46% base rate of positive outcomes. This is suggestive, but the n is tiny and the 60-session periods overlap the “falling 2y” episodes above (`out/rate_beta_onsets.csv`).

What sub-question 2 shows

- **The current value is extreme on two measures.** KRE’s rate beta is −0.062: the 3.7th percentile since 2006 and the 4.2th since 2012. Standardized (beta × σΔy / σr), it is −0.33, the **1.5th percentile**. The KRE SPY beta in the same model is 0.24, the 1.3th percentile.
- **The sign is not new.**
  - A negative beta was common in 2006–2011 (46–73% of days per year), but small in size. In the GFC/ZIRP era, falling yields came with bank stress.
  - It was **never negative in 2013–2021**. That decade produced the “+0.07 median” in the brief.
  - It returned in late 2022 and has been the majority state since **August 2023** (57% of days; 58% of 2024; 63% of 2026 so far).
  - The record low (−0.157) came in **May 2024**, after SVB. In that regime, higher yields meant bigger unrealized securities losses and funding pressure, so regionals traded like long-duration assets.
- **It is a bank-wide regime.** KBWB’s rate beta is also negative (−0.031, 4.4th percentile), and the two rolling series correlate at **0.98**. KRE is somewhat more negative: the KRE − KBWB beta gap is −0.031, the 8th percentile. That is consistent with the decoupling from study 1, but it is a difference of degree.
- **Hiking cycles.**
  - 2015–18: the beta was **always positive** (median +0.12). Higher rates meant NIM expansion.
  - 2022–23: median +0.02, negative on 19% of days. It turned negative from Sept 2022, as AOCI losses grew.
  - **SVB itself (Mar–May 2023): the beta was positive (+0.13).** KRE crashed while yields fell, which is the opposite of today. A negative rate beta is therefore *not* a crisis signature. In the acute 2023 bank run, KRE fell on falling yields.
  - Overall, P(beta < 0 | hiking) = 8% vs 33% outside hiking. **A strongly negative beta during a hiking phase, as in 2026, has almost no precedent in this sample.** The closest is late 2022 (−0.03 at worst).
- **Does a negative beta precede drawdowns?** On the non-overlapping 63-session grid, b < 0 (n=20, 14 of them from 2006–2012) was followed by:
  - forward-60 KRE of −0.9%, against a +2.0% base (t −1.1);
  - KRE − IWM of −3.6% (t −1.9);
  - a forward max drawdown of −14.3%, against a −13.2% base: no difference.

  With b < −0.03 (n = 8, mostly GFC), forward-60 is −8.6% and the drawdown −20%, but that sample is too thin and too GFC-heavy to use. **Verdict: a weak tendency toward relative underperformance, and no evidence of a larger drawdown.**

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3. Where 10/14–10/30/2026 sits

Table 3a — Current windows under the same rules

Window	Sessions	Δ2y	Δ10y	Δ2s10s	KRE	KBWB	IWM	SPY	KRE−KBWB	ΔBaa−10y	HYG−IEF	ΔHY OAS	KRE rate beta at start
2026-10-02 → 2026-10-30	20	-27bp	-34bp	-7bp	+1.7%	+1.0%	-0.8%	-1.8%	+0.8pp	+2bp	-2.4pp	+34bp	+0.003
2026-08-03 → 2026-10-30	63	+31bp	+24bp	-7bp	-6.0%	-8.4%	-5.5%	-0.0%	+2.4pp	-12bp	-0.9pp	+66bp	-0.124
2026-10-14 → 2026-10-30	12	-26bp	-25bp	+1bp	+5.1%	+3.4%	+2.9%	-1.8%	+1.6pp	+5bp	-1.5pp	+19bp	-0.008
2026-08-03 → 2026-10-14	51	+57bp	+49bp	-8bp	-10.5%	-11.5%	-8.2%	+1.8%	+0.9pp	-17bp	+0.5pp	+47bp	-0.124

Real vs breakeven split (FRED DFII10, T10YIE).

Period	10y nominal	Real 10y	10y breakeven
8/3 → 10/14	+49bp	+41bp	+8bp
10/14 → 10/29	−21bp	**−21bp**	**0bp**
10/22	−17bp	−12bp	−5bp

The rally in Treasuries was a fall in real yields and term premium, not lower expected inflation. A “constructive” easing story, where inflation cools and the Fed can stop, would normally show falling breakevens and a falling front end on Fed repricing. Here the 2-year did fall 26bp, but breakevens did not move.

Table 3b — Pattern scan: 12-session windows like 10/14→10/30

Criteria: 10y ≤ −15bp, KRE ≥ +3%, and *both* credit proxies widening. Clusters separated by more than 63 sessions; first date shown. The rule was written to match the current episode before its forward returns were printed. **This is exploratory.**

12-session window	Δ2y	Δ10y	KRE	KRE−KBWB	KRE−IWM	SPY	ΔBaa	HYG−IEF	ΔHY OAS	KRE rate beta	KRE F20	KRE F60	F60 vs KBWB	F60 vs IWM
2007-08-08 → 2007-08-24	-34	-21	+3.7	—	+3.5	-1.0	+14	-0.8	—	+0.001	-2.1	-16.9	—	-10.8
2008-01-04 → 2008-01-23	-65	-37	+5.1	—	+9.4	-5.3	+36	-4.4	—	+0.012	-0.2	-1.5	—	-5.5
2008-07-22 → 2008-08-07	-31	-22	+4.8	—	+5.1	-0.4	+12	-3.0	—	+0.018	+7.5	+5.3	—	+29.7
2008-11-18 →	-22	-86	+3.1	—	+0.7	+1.0	+36	-9.4	—	+0.008	-0.6	-46.6	—	-23.2

2008-12-05														
2009-03-13 → 2009-03-31	-17	-18	+7.2	—	+0.0	+5.3	+18	-0.7	—	-0.060	+12.1	-2.8	—	-24.2
2009-06-24 → 2009-07-13	-27	-34	+3.2	—	+3.6	-0.0	+18	-3.2	—	-0.007	+26.3	+14.6	—	-7.7
2009-11-04 → 2009-11-20	-16	-21	+6.5	—	+2.3	+4.3	+4	-0.6	—	-0.036	+6.4	+15.5	—	+7.5
2010-09-17 → 2010-10-05	-7	-25	+4.1	—	-1.8	+3.2	+8	-1.3	—	+0.015	-4.2	+15.2	—	+0.3
2014-02-12 → 2014-03-03	-3	-20	+3.0	+3.1	-0.9	+1.6	+1	-0.1	—	+0.020	+4.8	-2.9	-3.8	+0.2
2015-03-04 → 2015-03-20	-6	-19	+3.7	+1.5	+0.7	+0.5	+5	-2.3	—	+0.083	-0.2	+8.7	+1.4	+8.1
2015-11-12 → 2015-12-01	+2	-17	+3.1	+0.2	-1.3	+2.9	+2	-1.0	—	+0.165	-6.9	-22.9	-2.0	-9.0
2022-06-16 → 2022-07-06	-17	-35	+4.1	+2.1	-0.6	+5.0	+26	-2.5	—	+0.010	+8.4	+1.1	+5.1	+3.7
2023-12-08 → 2023-12-27	-51	-44	+9.3	+0.4	-0.7	+4.0	+3	-0.3	-41	-0.065	-1.9	-7.9	-13.9	-8.5
2024-04-23 → 2024-05-09	-6	-16	+3.5	+0.7	-0.2	+2.9	+2	-0.4	-6	-0.150	-6.3	+5.0	+5.3	+5.4
2025-01-10 → 2025-01-29	-19	-22	+9.1	-0.7	+4.8	+3.7	+3	-0.3	-13	+0.054	-3.7	-14.6	+0.1	-0.6
2025-08-18 → 2025-09-04	-18	-17	+6.8	+1.6	+3.0	+0.9	+10	-0.5	-4	-0.023	-4.4	-4.2	-6.6	-9.6
2026-01-27 → 2026-02-12	-6	-15	+3.6	+3.8	+5.5	-2.0	+7	-1.2	+21	-0.042	-10.7	-2.5	-1.8	-12.6
2026-06-08 → 2026-06-25	-6	-16	+6.9	+0.5	+1.4	-0.4	+2	-0.9	+3	-0.112	+1.3	-3.2	-1.2	+1.0
2026-10-07 → 2026-10-23	-8	-28	+5.2	+4.1	+5.9	-2.2	+1	-2.4	+29	-0.066	—	—	—	—
All 18 with F60											+1.4 (39%↑)	-3.4 (39%↑)	-1.7 (n=10)	-3.1 (44%↑)
Post-2012 (10)											-2.0 (30%↑)	-4.3 (30%↑)	-1.7 (40%↑)	-2.2 (50%↑)

(Reference: the mean 60-session KRE return over all start dates is +2.0%, positive 58% of the time.)

Placement.

- **Classification.**
  - *20-session window (10/2 → 10/30):* falling 2y (−27bp) and 10y (−34bp), KRE up (+1.7%), KRE beat KBWB (+0.8pp), Baa wider (+2bp), HYG < IEF (−2.4pp), HY OAS +34bp, 2s10s −7bp. That is **STRESS REBOUND, with bull flattening and KRE outperforming KBWB.**
  - *63-session window (8/3 → 10/30):* **not** a falling-yield window. Both yields rose (2y +31bp, 10y +24bp). The rally is a partial retracement inside a rising-yield quarter.
- **10/22 is close to unique.** Only one other day since 2006 had the 10-year down ≥10bp, SPY down ≥1% and KRE up ≥1.5%: **1/22/2008**, the day of the Fed's emergency 75bp cut, whose effective date in FRED DFEDTAR is 2008-01-22. More broadly, on days when the 10-year fell ≥10bp, KRE rose on **0% of such days in most years from 2012 to 2021**, against 47–100% since 2022. Regionals rallying on a bond rally is a post-2022 behaviour.
- **Which episodes it resembles most, and why:**
  1. **Post-2023 "AOCI-relief" rallies: Dec 2023, May 2024, Jan 2025, Sep 2025, Feb 2026, Jun 2026.** The mechanism is the same: KRE rallies *because* yields fall. The rate beta was already negative in five of the six; Jan 2025 is the exception.
     - Forward-60 KRE returns: −7.9%, **+5.0%**, −14.6%, −4.2%, −2.5% and −3.2%. That is 5 of 6 negative, mean −4.6%.
     - Four of the six lagged KBWB, Dec 2023 by −13.9pp. May 2024, the most negative-beta case (b −0.150), is the counter-example: +5.0% absolute and +5.3pp against KBWB.
     - Jan 2025 ran into the tariff shock, so it is not a clean read.
  2. **Late-2007 / Jan-2008 flight-to-quality rebounds.** These had the same *market* signature: falling yields, wider credit, KRE up while SPY fell, and in Jan 2008 the 10/22-type day itself. Forward-60 returns were −16.9% and −1.5%. The difference is important: then, bank credit was the source of the stress. Now the stress comes from outside banking (AI capex, France, the Middle East). Note that HY OAS at 3.44 is *near the top* of its FRED-available 2023–26 range (2.59–4.61); it was wider than on 88% of days since Nov 2023. Pre-2023 levels cannot be checked from FRED's free series, so I cannot say whether 3.44 is high or low in long-run terms.
  3. **Not like SVB (Mar 2023).** Then, yields fell *and* KRE crashed, and the rate beta was positive.
  4. **Not like 2015–18 hiking.** Then, regionals rose *with* yields.
- **The current episode also differs from every analog in one way.** The Fed is *hiking*: +25bp on 9/16/2026, a hold with two hike dissents on 10/28, and a December hike reportedly in play. In earlier falling-yield rallies the Fed was cutting or on hold. A hiking Fed caps how far the front end can fall, so the rate-relief channel is more fragile.

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4. Cross-section: which traits matter on falling- vs rising-yield days (optional study)

- **Universe and data.** The 167 KRE holdings (SSGA file dated 30-Oct-2026) were mapped to 6/30/2026 FDIC call reports. Charters were summed by holding company (Wintrust = 17 charters in the 6/30 data; PNFP includes the former Synovus charter).
- **Mapping corrections.** 14 tickers needed manual mapping because the automatic match picked a different bank with the same holding-company name: FNB, CFG, LOB, THFF, CCNE, PEBO, IBCP, HTH, TFSL, CLBK, AMAL, MCHB, FLG, TOWN (`xs/overrides.csv`).
- **Returns.** Excess return = stock minus the equal-weighted 167, from 7/1 to 10/30/2026.
- **Days.** Falling days are Δ10y ≤ −5bp (8 days, including 10/22); rising days are Δ10y ≥ +5bp (14 days).
- **Model.** OLS with HC3 errors; each trait is z-scored and size-controlled.

Table 4 — Coefficients per 1 SD of each trait (bp of daily excess return), HC3 t in brackets

Trait (1 SD; size-controlled)	Falling days: coef bp (t)	Rising days: coef bp (t)	Fall − rise (t)	Relative rate beta: ×1e4 (t)	Rate-beta t, robustness: drop 10/22 · ±3bp · Apr–Oct
Unrealized securities gain(+)/loss(−) ÷ T1	-2.3 (-0.9)	-3.6 (-2.2)	+0.4	-39 (-2.0)	-2.4 · -2.0 · -2.3
HTM unrealized gain/loss ÷ T1	-5.8 (-2.2)	-2.6 (-1.6)	-1.0	+1 (+0.1)	-1.3 · +0.1 · +0.1
HTM share of securities	+5.4 (+1.7)	-0.3 (-0.2)	+1.2	-49 (-1.4)	— · — · —
Securities ÷ assets	+2.9 (+0.8)	+3.7 (+1.6)	-0.2	+36 (+1.6)	— · — · —
Deposit cost (H1 annualized)	+0.8 (+0.2)	-4.2 (-2.1)	+1.0	-45 (-1.5)	-1.0 · -1.5 · -2.2
Uninsured ÷ deposits	+1.1 (+0.3)	+3.4 (+1.7)	-0.6	+14 (+0.5)	— · — · —
Loans ÷ deposits	-0.2 (-0.1)	-5.5 (-2.7)	+1.0	-46 (-1.6)	-1.5 · -1.6 · -2.0
CRE concentration (÷ capital)	+2.6 (+0.8)	-5.2 (-2.3)	+1.8	-69 (-2.7)	-1.9 · -2.7 · -2.9
log assets	+9.6 (+3.2)	-5.2 (-3.1)	+3.9	-56 (-2.8)	-4.0 · -2.8 · -2.5

What sub-question 4 shows

- **Size is the rate exposure inside KRE.** Larger KRE banks gained more on falling-yield days (+9.6bp per SD, t 3.2) and lost more on rising-yield days (−5.2bp, t −3.1). Their relative rate beta is more negative (t −2.5 to −4.0 across the robustness variants). This matches study 1's "size = beta" finding and points to the same channel that makes KBWB's beta negative.
- **CRE concentration, loan/deposit ratio and deposit cost predict *rising-yield-day* losses.** The t-values are −2.1 to −2.7, and the rate-beta t for CRE is −1.9 to −2.9 across variants. The within-KRE rate channel looks like **credit and funding** ("higher for longer hurts CRE borrowers and deposit costs"), not securities marks.
- **Securities losses work the "wrong" way for an AOCI story.** Larger unrealized losses relative to T1 come with a *less* negative relative rate beta (t ≈ −2.0 to −2.4, where a negative coefficient means bigger losses give a higher beta).
- **HTM loss / T1** was associated with better returns on falling days (t −2.2, same sign convention) *and* on rising days (t −1.6). The fall-minus-rise difference is insignificant (t −1.0). **The prior "bigger HTM losses → bigger rally" result is better read as general outperformance of those names than as rate relief.**
- **Caveats.** 8–22 event days; 9 traits × 4 outcomes; earnings-season days fall inside the window. The robustness variants were dropping 10/22, using a ±3bp threshold, and using an Apr–Oct window. Across them, size (|t| 2.5–4.0), securities loss (|t| 2.0–2.4) and CRE (|t| 1.9–2.9) keep their sign and size for the relative rate beta, while deposit cost and loan/deposit weaken. Re-run on 9/30 call reports when FDIC loads them.

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5. Confidence & gaps

Where I'm fairly confident

- The re-verified market and macro facts (§0). The Fed actions are confirmed in Fed press releases.
- The rate beta is at an extreme (1.5th–4th percentile). It is shared with KBWB, so it is not KRE-specific. A negative sign has been the norm since Aug 2023, but it is rare during a hiking phase.
- Falling yields, and KRE rallies within them, have been followed by KRE *lagging KBWB* over ~60 sessions. This holds on both grids for the 2-year, survives the alternative thresholds and the COVID/SVB exclusion, and the median agrees.

Where I'm not confident

- **Whether today's rally is "constructive" or a "temporary rebound."** The pre-defined credit split does not separate outcomes robustly. Cells are 5–12 windows; results flip with the grid phase (Table 1f); the alternative thresholds leave 1–3 cases; and 2008 dominates the stress-rebound cells.
- **The credit measure itself.** HY OAS history is unavailable from FRED's free series, and the two proxies agree with HY OAS on direction only about 70% of the time. A licensed full ICE BofA HY OAS history, or ALFRED once it is reachable, could re-classify a quarter of the windows. **This is the single change most likely to alter sub-question 1's credit results.**
- **The analog scan (Table 3b)** is exploratory. The rule was chosen to resemble the current episode, the forward periods include exogenous shocks, and 7 of 18 cases are from 2007–09.
- **The cross-section** rests on 8 falling days, and the FDIC mapping relies on name matching with 14 manual fixes. A mis-mapped bank would add noise rather than bias.

Where sources or data disagreed

- The brief's SPY beta (0.21) and R^2 (0.17) differ from mine (0.24, 0.15). The rate coefficient matches (−0.061 vs −0.062). The likely cause is the yield source or window alignment; it does not change any conclusion.
- The brief's "historical median +0.070" is a since-2012 median. The full-history (2006) median is +0.042.
- The press gives slightly different 10-year closes for 10/22 (CNBC 4.972%; Treasury par 4.96%). I use Treasury's par curve throughout.

Claims resting only on secondary sources

- That futures price a December hike (CNBC).
- The names of the 10/28 dissenters (CNBC, J.P. Morgan AM; the Fed statement in hand shows only the 10–2 vote).
- The narrative causes of 10/22: the AI selloff, French fiscal stress and Mideast tensions (CNBC, Bloomberg).
- The market data for 10/22 *is* primary (Treasury, Yahoo prices).

What would change the conclusion

1. Full HY OAS history re-classifying the credit cells.
2. Breakevens and the front end starting to fall *together*, with HY OAS tightening, which would move the episode toward the CONSTRUCTIVE cell.
3. The rate beta turning positive again (yields rising while KRE rises), which would mark a return to the NIM regime.
4. 9/30 call reports showing deposit-cost or CRE deterioration among the names that led the rally.
5. For study 4: this study suggests that low correlation plus a negative rate beta overlaps heavily with "KRE rallies on falling yields". Any low-correlation predictive test should control for that regime.

Thin cells (explicit list).

- All W=20 stress-rebound and benign cells at the 60-session horizon (n = 2–8).
- All alternative-threshold sub-cells (n = 0–7).
- Every rate-beta condition with b < −0.03 (n = 4–8).
- All "vs KBWB" cells before 2012 (no data).

Appendix A — Files (all in this folder)

File	What it is
`REPORT.md` / `REPORT.pdf`	this report
`PREREGISTERED_SPEC.md`	classification rules fixed before forward returns, with a change log
`build_data.py`	downloads → `data/master.csv` (KRE, KBWB, IWM, SPY, HYG, IEF adj. closes; Treasury 2y/10y/3m; HY OAS; BAA10Y)
`regime.py`	sub-question 1 → `out/windows_W20.csv`, `out/windows_W63.csv`, `out/regime_table.csv` (all cells × thresholds × horizons × outcomes, with n, episodes, mean, median, hit, baseline, diff, t), `out/W20_h60_phase_robustness.csv`
`ratebeta.py`	sub-question 2 → `out/rolling_rate_beta.csv`, `out/rate_beta_episodes.csv`, `out/rate_beta_forward*.csv`, `out/rate_beta_onsets.csv`, `out/rate_beta_chart.png`
`current.py`	sub-question 3 → `out/current_classification.csv`, `out/analogs_12d.csv`, `out/analogs_nn_*.csv`, `out/days_like_1022.csv`, `out/real_vs_breakeven.csv`
`explore_ratebeta_interaction.py`	exploratory: rate-beta sign × falling yields → `out/exploratory_ratebeta_interaction.csv` (all cells n ≤ 10; inconclusive)
`regime_map.py`	`out/regime_map.png`
`xs/match.py`, `xs/overrides.csv`, `xs/mapping.csv`	KRE ticker → FDIC holding-company mapping
`xs/crosssection.py`	sub-question 4 → `xs/traits.csv`, `xs/traits_and_outcomes.csv`, `xs/xs_results.csv`, `xs/xs_robustness.csv`
`make_tables.py`	builds this report's tables from the CSVs

To reproduce: re-download the raw files with the URLs below into `data/` (Yahoo JSON per ticker into `data/` and `data/px/`), then run `build_data.py → regime.py → ratebeta.py → current.py → xs/match.py → xs/crosssection.py → make_tables.py`.

Appendix B — Sources

- **US Treasury, daily par yield curve CSVs, 2005–2026:** `https://home.treasury.gov/resource-center/data-chart-center/interest-rates/daily-treasury-rates.csv/{YEAR}/all?type=daily_treasury_yield_curve&field_tdr_date_value={YEAR}&page&_format=csv`
- **FRED series:** BAMLH0A0HYM2 (truncation note on the FRED series page), BAA10Y, DFEDTAR, DFEDTARU, DFII10, T10YIE, NFCI (downloaded, not used). Pattern: `https://fred.stlouisfed.org/graph/fredgraph.csv?id=…`
- **Federal Reserve press releases:**
  - FOMC statement, Sept 16, 2026: https://www.federalreserve.gov/newsevents/pressreleases/monetary20260916a.htm
  - FOMC statement, Oct 28, 2026: https://www.federalreserve.gov/newsevents/pressreleases/monetary20261028a.htm
  - Minutes, Sept 15–16, 2026: https://www.federalreserve.gov/monetarypolicy/fomcminutes20260916.htm
- **FDIC BankFind financials API** (REPDTE 20260630): https://api.fdic.gov/banks/financials
- **SSGA KRE daily holdings** (as of 30-Oct-2026): https://www.ssga.com/library-content/products/fund-data/etfs/us/holdings-daily-us-en-kre.xlsx
- **Yahoo Finance chart API v8** (adjclose; explicit period1/period2) for ETF and stock prices. This is a secondary data vendor for prices; there is no free primary source for adjusted ETF prices.
- **Secondary (named where used):**
  - CNBC, Oct 28 2026 Fed coverage (dissenters; December pricing);
  - CNBC / Bloomberg, Oct 22 2026 market coverage (drivers of the Treasury rally);
  - J.P. Morgan AM, FOMC Oct 2026 note (dissenters).